

Customer	Thanh Thien
Instrument	METONE BAM1020
ID No.	13-0551
System/Job No.	500926

BAM1020 Calibration Report

Notes:

Calibration Performed by	Khanh Nguyen
Date	28-08-13
Time Begin/End	12:00 PM
Location	Vinh Yen

Calibration Equipment

Test Equipment	ID #	Calibration Due Date	
Temperature Probe ID TE	330	10-11-13	PASS
Barometer ID TE	328	08-09-13	PASS
Flow Standard TE*	285	07-09-13	PASS

* Note: If using a DryCal unit with separate base and flow cells, you must record the ID's of both parts

Displayed Instrument Parameters

Inlet Head Type PM10

Date and Time Correct Y

Station #	1
Range (1.000mg)	1
BAM Sample time	50
MET Sample time	60
OFFSET (-0.015mg)	-0.015
Flow Rate (16.7 Lpm)	16.7
Conc Type	STD
Flow Type	ACTUAL
Cv	0.96
ABS	0.825

K Factor	0.951
STD Temp. (0°C)	0
Qo	-0.096
*u sw	0.301
*BKGD	-0.0011
Heater Control	AUTO
e1	-0.015
e3	0.5
e4	0
Errors	15

AP	150
FRI	10
FRh	20%
Cycle Mode	STANDARD
RH Control	YES
RH Setpoint (35%)	35%
Datalog RH	YES
Delta T Control	NO
Delta T Setpoint	99
Datalog Delta T	NO
Firmware Rev	81236-02 V1.0.0

Note: Where possible, Concentration type should be set to Standard and Flow type should be set to Actual.

Real-Time Module check (2-monthly)

Real Time Module Installed?

NO

Flow rate is not critical but should be maintained. Record the initial flow rate and adjust if required.

	Nominal Flow	Initial Flow	Final Flow
Flow Check	2.0 lpm		

Leak Check and Flow Audit (3 monthly)

Pre-adjustment check

	Displayed	Reference	Error	PASS / FAIL
Ambient Temperature (°C)	31.3	31.0	-0.3	PASS
Barometric Pressure (mmHg)	752.0	751.8	-0.22	PASS
Flow Rate Actual (Lpm)	16.7	16.185	-3.2%	Calibrate

Check filter tape for clean, even spots with sharp edges. Note any holes in spots.

Initial Leak Check	0.3	PASS
Final Leak Check	0.1	PASS

Clean nozzle and vane where necessary

Displayed flow during leak check must be <1.0 Lpm

All Flow Calibrations must be performed in Actual mode, record all results before making any adjustments

If the flow rate error is greater than 6% then data must be invalidated.

If the temperature or pressure sensor check fails, adjust the calibration and record the final results.

The flow rate should be re-calibrated if greater than 2% error. Complete the 12-monthly multi-point flow calibration if required.

Post adjustment Check

	Displayed	Reference	Error	PASS / FAIL
Ambient Temperature (°C)	31.3	31.0	-0.3	PASS
Barometric Pressure (mmHg)	752.0	751.8	-0.22	PASS
Flow Rate Actual (Lpm)	16.7	16.634	-0.4%	PASS

Flow Calibration (12 monthly)

	Displayed	Reference	Error	PASS / FAIL
Ambient Temperature (°C)	31.3	31.0	-0.3	PASS
Barometric Pressure (mmHg)	752.0	751.8	-0.22	PASS
Flow 1 (15 lpm)	15	14.984	Pump Check	PASS
Flow 2 (18.4 lpm)	18.4	18.3		
Flow 3 (16.7 lpm)	16.185	16.634		

Note: If pump check fails then repair or replace the pump

Post Calibration flow Audit

Set Flow (Lpm)	Displayed	Reference	PASS / FAIL
16.7	16.7	16.8	PASS

Must be $\pm 2\%$

Filter RH and Filter Temp Calibration Check

Note: Allow sufficient time for the sensors to equilibrate before making any adjustment.

Initial

	Displayed	Reference	PASS/FAIL
Filter T (°C)	38.9	38.0	PASS
Filter RH (%)	21%	25%	PASS

Must be $\pm 1^{\circ}\text{C}$
Must be $\pm 4\%RH$

Final

	Displayed	Reference	PASS / FAIL
Filter T (°C)	38.9	38.0	PASS
Filter RH (%)	21%	25%	PASS

Must be $\pm 1^{\circ}\text{C}$
Must be $\pm 4\%RH$

Membrane Calibration Check (12-monthly)

	ABS (mg)	Last M	Error (%)	PASS/FAIL
Internal Membrane	0.825	0.824	0.1%	PASS

Note: If the membrane check fails, inspect the membrane for dust/damage and repeat test.

Technician Signature		Date	
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